

AI DM Conversation Agent — Architecture Documentation

Purpose

AI-driven DM agent that answers questions from a knowledge base and transfers the conversation to a human agent when requested.

Core Capabilities

- AI conversation responses
- Knowledge base retrieval
- Intent detection
- Human handoff
- REST API integration

Backend Stack

Runtime: Python 3.11

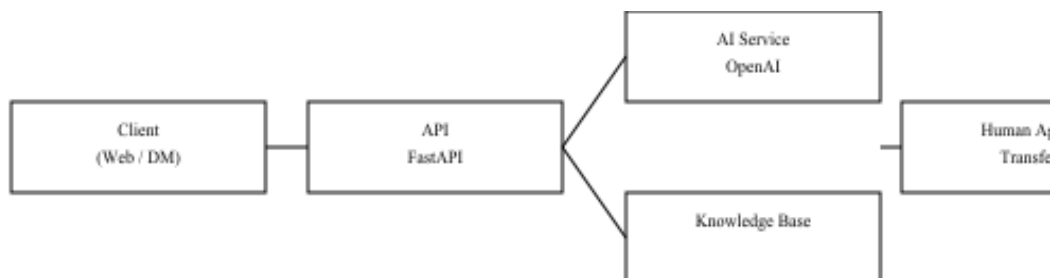
Framework: FastAPI

Server: Uvicorn

Environment: python-dotenv

AI: OpenAI API

System Architecture



Request Flow

Step	Component	Action
1	Client	Send POST /chat
2	API Layer	Receive request
3	Intent Logic	Detect knowledge / handoff
4	AI Service	Generate response
5	API	Return answer

Project Structure

project-root



main.py

requirements.txt

.env



routes

chat.py



services

ai_agent.py

knowledge_service.py



controllers

chat_controller.py



utils

logger.py

Future Extensions

- Vector database for knowledge retrieval
- Worker queue for message processing
- Conversation memory storage
- Analytics dashboard
- Multi-channel integrations (Instagram, WhatsApp, Web chat)